

Chapter 2

Applications

Sensitivity analysis and uncertainty quantification are required for most large-scale engineering, physical, and biological applications. Furthermore, the complexity of models often exacerbates associated challenges. We illustrate in this chapter issues pertaining to sensitivity analysis and uncertainty quantification for models employed for nuclear power plant design, quantitative systems pharmacology, weather, and climate. We also illustrate roles of uncertainty analysis when constructing digital twins and virtual populations.

In later chapters, we also employ the following examples to illustrate specific topics associated with additional large-scale applications:

- Bayesian inference and uncertainty quantification for a wetland methane emission model: Chapters 12 and 13;
- Surrogate model construction for MCNP simulations employed to quantify radiation transport in an urban environment: Chapter 16;
- Gaussian process emulator construction for volcanic flows: Chapter 18;
- Reduced-order model construction for a chemical vapor deposition reactor, turbulent flow over an airfoil, and the use of greedy sampling algorithms to facilitate uncertainty propagation for a thermal fin: Chapter 19.

Whereas these examples illustrate a range of phenomena, they omit numerous applications where uncertainty quantification serves a critical role. We refer readers to the literature in their respective fields for details regarding applications which require uncertainty quantification.

2.1 Nuclear Power Plant Design

Nuclear power plants presently provide approximately 19% of the electricity in the United States and 15% of electricity worldwide. Hence they constitute a major source of sustainable energy. Whereas the efficiency of nuclear power plants has improved over the last 40 years, the majority of operating plants have retained the original fundamental designs. This has motivated significant research regarding the

development of simulation-based design tools to improve reactor efficiency, reduce operating costs, reduce nuclear waste, and continue to enhance safety. The broad objective is to improve the operation of current nuclear power plants and guide the design of next generation reactors.

We summarize here fundamental aspects of the reactor physics and models to motivate the central role of sensitivity analysis and uncertainty quantification. This will provide examples, which we will employ in subsequent chapters, to illustrate the end-to-end process of uncertainty quantification for this large-scale application.

2.1.1 Light Water Reactors

Heat in nuclear reactors is produced by controlled nuclear fission in the reactor core. The core contains nuclear fuel rods, control rods, and water-filled channels. The fuel rods, which are approximately 12 feet long, contain uranium or uranium oxide pellets. The control rods contain elements such as cadmium that absorb neutrons, thus slowing chain reactions. We focus on light water reactors (LWR), which employ ordinary water – termed light water – as a coolant and neutron moderator [119,339].

We illustrate a pressurized-water reactor (PWR) in Figure 2.1. In this design, coolant is circulated through the core under high pressure so it remains liquid despite being raised to temperatures on the order of 315°C . This hot primary coolant then flows through a heat exchanger, where it generates steam in an isolated secondary coolant that in turn drives turbines. The separation of the two coolant sources prevents accidental radioactive contamination of the secondary coolant source if there is an accidental breach in the primary system.

There are two primary mechanisms for controlling the power generated by a PWR: insertion or retraction of the neutron absorbing control rods and varying the concentration of boric acid dissolved in the reactor coolant. The latter acts in a manner similar to the control rods since boron readily absorbs neutrons.

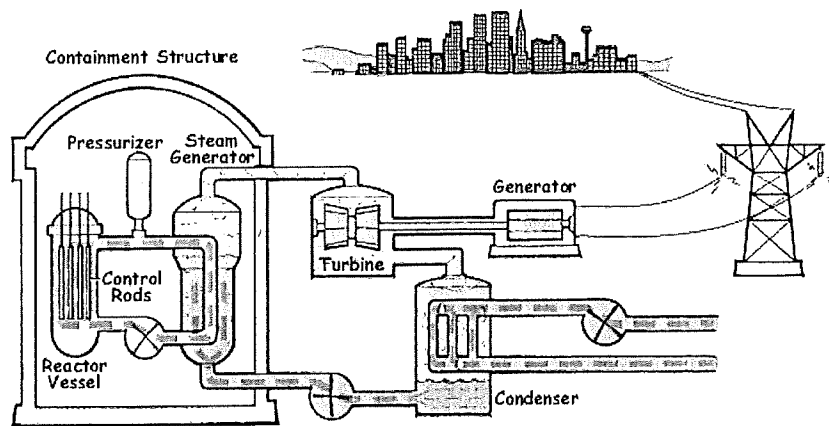


Figure 2.1. Schematic of pressurized-water reactor (PWR). Image courtesy of United States Nuclear Regulatory Commission.

2.1.2 Crud-Induced Phenomenon

During reactor operation, metal oxide corrosion products, such as nickel ferrite (NiFe_2O_4), accumulate on cladding surfaces at the outer layer of fuel rods to form what is termed *crud*. Within this porous crud layer, boron employed in the coolant as a reaction moderator precipitates to form lithium tetraborate ($\text{Li}_2\text{B}_4\text{O}_7$). Because boron absorbs neutrons, this locally suppresses power and can produce a crud-induced power shift (CIPS) in the nuclear fuel. Due to its significant impact on reactor performance, the prediction and assessment of crud and CIPS constitutes a long-standing challenge in the nuclear industry.

2.1.3 Reactor Models

The physics in the core of a PWR involves multiscale interactions between neutron transport, thermal-hydraulics, the chemistry associated with the use of boron in the coolant to control reactions, and properties of the fuels. The coupled multiphysics nature of the application constitutes one of the significant challenges for uncertainty quantification. We summarize aspects of the neutronics, thermal-hydraulics, and chemistry models to motivate some of the issues which we address in subsequent chapters. We illustrate in Example 2.4 the construction of a digital twin for nuclear power plant prognosis and operation assessment.

Neutron Transport Equations

The quantification of neutron distributions in the reactor core is central to reactor models since neutron densities and energy levels govern the nuclear reactions in the reactor. The neutron interaction with the primary coolant is also critical since the coolant serves as a neutron moderator and the neutrons heat the coolant.

To specify the neutron distribution, one typically considers the angular neutron flux $\varphi(\mathbf{r}, E, \boldsymbol{\Omega}, t)$, where $\mathbf{r} = (x, y, z)$ is a position vector, E is the energy level, and $\boldsymbol{\Omega} = \mathbf{v}/|\mathbf{v}|$ is a solid angle, which specifies the direction of motion in terms of the velocity $\mathbf{v}$.

To model neutron processes in the reactor core, one balances neutron sources and loss mechanisms for an arbitrary volume V . Sources include fission, neutrons entering V , and neutrons which incur a scattering collision $E' \rightarrow E, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}$ from energies and directions $E, \boldsymbol{\Omega}$ to $E', \boldsymbol{\Omega}'$. Losses are due to neutrons leaving V and neutrons incurring a collision.

As detailed in [119], a balance of the source and loss terms yields the 3-D neutron transport equation

$$\begin{aligned} \frac{1}{|\mathbf{v}|} \frac{\partial \varphi}{\partial t} + \boldsymbol{\Omega} \cdot \nabla \varphi + \Sigma_t(\mathbf{r}, E) \varphi(\mathbf{r}, E, \boldsymbol{\Omega}, t) \\ = \int_{4\pi} d\boldsymbol{\Omega}' \int_0^\infty dE' \Sigma_s(E' \rightarrow E, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}) \varphi(\mathbf{r}, E', \boldsymbol{\Omega}', t) \\ + \frac{\chi(E)}{4\pi} \int_{4\pi} d\boldsymbol{\Omega}' \int_0^\infty dE' \nu(E') \Sigma_f(E') \varphi(\mathbf{r}, E', \boldsymbol{\Omega}', t), \end{aligned} \quad (2.1)$$

where Σ_t, Σ_f are macroscopic total and fission cross-sections, which are defined as the ratio of the reaction rate (s^{-1}) to incident flux ($\text{cm}^{-2}\text{s}^{-1}$) when a beam of particles impinges on a nucleus. The scattering cross-section Σ_s characterizes scatter from (E', Ω') to (E, Ω) in the cone $d\Omega$. Finally, $\chi(E)$ and $\nu(E')$ respectively denote the fission spectrum and average number of fission neutrons produced by fission resulting from neutrons with energy E' . There exist a number of commercial and government laboratory neutronics codes. We illustrate aspects of SCALE6.1 in later chapters.

Thermal-Hydraulic Models

Models for the primary coolant must incorporate fluid and gas phases, heat transfer, and neutron interactions. We summarize aspects of two-phase thermal-hydraulic models and refer readers to [74, 188, 310] for details regarding their derivation.

Let α_g, α_f respectively denote the volume fraction of gas and fluid and let ρ_g, ρ_f and v_g, v_f denote the densities and velocities of the gas and fluid phases. The internal energies for the two phases are denoted by e_g, e_f . As detailed in (116)–(123) of Chapter 16 of [74], conservation of mass, momentum, and energy respectively yield the fluid-phase relations

$$\begin{aligned} \frac{\partial}{\partial t}(\alpha_f \rho_f) + \nabla \cdot (\alpha_f \rho_f v_f) &= -\Gamma, \\ \alpha_f \rho_f \frac{\partial v_f}{\partial t} + \alpha_f \rho_f v_f \cdot \nabla v_f + \nabla \cdot \sigma_f^R + \alpha_f \nabla \cdot \sigma + \alpha_f \nabla p_f \\ &= -F^R - F + \Gamma(v_f - v_g)/2 + \alpha_f \rho_f g, \\ \frac{\partial}{\partial t}(\alpha_f \rho_f e_f) + \nabla \cdot (\alpha_f \rho_f e_f v_f + T h) &= (T_g - T_f)H + T_f \Delta_f \\ &\quad - T_g(H - \alpha_g \nabla \cdot h) + h \cdot \nabla T - \Gamma[e_f + T_f(s^* - s_f)] \\ &\quad - p_f \left(\frac{\partial \alpha_f}{\partial t} + \nabla \cdot (\alpha_f v_f) + \frac{\Gamma}{\rho_f} \right) \end{aligned} \quad (2.2)$$

with analogous coupled gas-phase relations.

Energy, momentum, and mass exchange at the vapor/liquid interfaces are respectively modeled with the constitutive relations

$$\begin{aligned} H &= \kappa(T_f - T_g), \\ F &= \zeta(v_f - v_g), \\ \Gamma &= \gamma[(s_f - s_g)(T_{sat}(p_f) - T_g) - K_f], \end{aligned} \quad (2.3)$$

where T_f, T_g and s_f, s_g are the temperature and entropy density of the two phases. Furthermore, $T_{sat}(p_f)$ is the saturation temperature at the continuous phase pressure p_f , and κ, ζ , and γ are positive transport coefficients, which must be estimated during model calibration. Further details regarding the derivation of the thermal-hydraulic equations and required constitutive relations can be found in [74, 189, 310].

In subsequent examples, we discuss the high-fidelity computational fluid dynamics (CFD) code STAR-CCM+ and the subchannel code COBRA-TF (CTF). The former can resolve fine-scale fluid physics but requires 10^2 to 10^3 core hours for simulations. By employing a subchannel spacing at the level of the finest lateral mesh, similar simulations can be performed in the order of 5 minutes on a single processor. However, the ramification of the coarse mesh is the loss of fine-scale physics, which affects rod temperatures and turbulent and boundary layer flow behavior due to spacer grids.

Clad Surface Chemistry

We noted in Section 2.1.2 that the accumulation of nickel ferrite, NiFe_2O_4 (crud), on cladding surfaces can produce crud-induced power shifts (CIPS). A full characterization of these processes involves complex interactions between material chemistry, heating, and mass transport. We summarize representative relations to illustrate aspects of the models and refer readers to [99] for a comprehensive discussion regarding models for clad surface chemistry.

The growth of the crud layer on the cladding surface is modeled by the kinetic equation

$$\frac{dc_{\text{NiFe}_2\text{O}_4}}{dt} = (k_{\text{nonboil}} + k_{\text{boil}}q_s)c_{\text{coolant}} - k_{\text{erosion}}. \quad (2.4)$$

Here $c_{\text{NiFe}_2\text{O}_4}$ denotes the nickel ferrite concentration, k_{nonboil} and k_{boil} are non-boiling and boiling deposition rates, q_s is the subnucleate boiling heat flux at the crud surface, c_{coolant} denotes the particulate concentration in the coolant, and k_{erosion} quantifies the decrease in growth rate due to erosion. From the perspective of uncertainty quantification, the parameters $\theta = [k_{\text{nonboil}}, k_{\text{boil}}, k_{\text{erosion}}]^T$ must be calibrated for specific designs and operating conditions.

The deposition of crud is highly dependent on the temperature distribution throughout the crud layer. This in turn depends on properties of "chimneys" to the cladding which yield high concentrations of dissolved species and increased local saturation temperatures.

At steady-state, the temperature T is modeled by the heat equation

$$\nabla \cdot k \nabla T = q_{cw},$$

where k denotes the thermal conductivity and q_{cw} notes the chimney wall heat flux. It is established in [99] that the heat flux can be expressed as

$$q_{cw} = h(T - T_{\text{sat}}), \quad (2.5)$$

where the heat transfer coefficient $h = 2\pi r_{\text{chim}} h_{\text{chim}} \rho_{\text{chim}}$ depends on the characteristic radius, boiling heat transfer coefficient, and surface density of a chimney. One aspect of uncertainty quantification, when modeling CIPS, focuses on calibrating and quantifying uncertainties associated with $\theta = [T_{\text{sat}}, h_{\text{chim}}, k]^T$.

2.1.4 Roles of Sensitivity Analysis and Uncertainty Quantification

We summarize here selected uncertainties and errors associated with nuclear reactor models, some of which we illustrate in later chapters.

- The inherently multiphysics nature of phenomena such as the crud-induced power shift (CIPS) discussed in Section 2.1.2 – which couples neutronics, thermal-hydraulics, and chemistry – constitutes a significant challenge when performing sensitivity analysis and uncertainty quantification. This generally requires the initial construction of surrogate models which incorporate the inherent multiphysics coupling. We illustrate in Example 18.10 the construction of a single-physics surrogate for COBRA-TF (CTF), which we employed for subsequent Bayesian inference.
- The models and simulation codes have a number of phenomenological parameters, which must be specified or calibrated for a specific design. This includes the parameters $\theta = [k_{nonboil}, k_{boil}, k_{erosion}]^T$ in the kinetic clad equation (2.4) and $\theta = [T_{sat}, h_{chim}, k]^T$ in the chemistry heat flux relation (2.5). Furthermore, many of these parameters cannot be uniquely specified, in the sense detailed in Chapter 7, based on observed data. We illustrate in Section 9.6.2 the use of sensitivity analysis to determine 3 influential parameters from among 33 in COBRA-TF. We subsequently illustrate in Example 13.12 the use of Bayesian inference to quantify uncertainties associated with these 3 parameters and the construction of a posterior predictive distribution for an observed model response.
- Discretization of energy groups, and the incorporation of relevant reactions, in neutronics codes can yield 10^3 – 10^5 inputs. We illustrate in Section 10.6 the use of active subspace techniques for SCALE6.1 to isolate a low-dimensional subspace for 7700 inputs.
- The thermal-hydraulics and chemistry models exhibit varying degrees of uncertainty in closure and constitutive relations used to characterize phenomena such as turbulence and void fractions. This model error or discrepancy can significantly diminish the accuracy of model predictions, especially for extrapolatory or out-of-data regimes. We illustrate the use of machine learning for a turbulence relation in Example 14.5.
- The thermal-hydraulics code COBRA-TF exhibits numerical errors since lateral meshes cannot be resolved more finely than the dimension of subchannels. This has necessitated the development of flow regime-dependent closure relations to quantify phenomena such as rod-to-fluid heat transfer.
- The harsh environment inside a nuclear reactor limits the availability of measurements which can be obtained for model calibration, validation, and experimental design. This has motivated the use of *validated* high-fidelity simulation codes to inform, improve, and calibrate low-fidelity codes use for design, performance analysis, and risk assessment [147, 223]. For example, this includes use of the high-fidelity CFD code STAR-CCM+ to inform COBRA-TF.

Workflow

These motivate the following workflow, which is typical for large-scale applications.

- Construct surrogate or reduced-order models for subsequent sensitivity and uncertainty analysis (Chapters 15–19).
- Employ the sensitivity analysis techniques in Chapters 8 and 9, or active subspace techniques in Chapter 10, to determine subsets or subspaces of parameters which are influential or identifiable in the sense established in Chapter 7.
- Employ frequentist or Bayesian techniques to quantify uncertainties for model parameters or observation errors (Chapters 11 and 12).
- Propagate input uncertainties through models to construct prediction intervals for responses or quantities of interest (Chapter 13).
- Employ prediction intervals and validation data to establish and address model discrepancy (Chapter 14).

2.1.5 References

The text [119] provides a comprehensive exposition of issues pertaining to nuclear reactor design and associated models. This can be complemented by the text [339]. We refer readers to [224] for details regarding computational techniques for the neutron transport equation and [189] for information regarding the thermal-hydraulic equations. The *Handbook of Nuclear Engineering* [74] details numerous aspects of the field. Specifically, readers are respectively referred to Chapters 1, 5, 7, 16, 17, and 28 for information regarding the principles of neutron transport, mathematics for nuclear engineering, multiphase flows, and sensitivity and uncertainty analysis.

2.2 Quantitative Systems Pharmacology (QSP) Models

Biological applications introduce a range of phenomena, which exacerbate both the challenges associated with developing accurate mechanistic models and the potential contributions of these models to predictive science. We illustrate here challenges associated with sensitivity analysis and uncertainty quantification in the context of quantitative systems pharmacology (QSP). QSP models broadly focus on quantifying mechanisms which determine the response of biological systems to drug regimens. Specifically, we focus on physiologically-based pharmacokinetic (PBPK) models, which quantify the processes that determine drug distribution, absorption, and elimination during a time-dependent drug regimen. One emphasis in PBPK models is to employ compartments associated with affected organs and represent, to the degree possible, the associated physiological processes in terms of equations and parameters having physiological interpretations and constraints. The latter is critical for yielding parameters which can be estimated using available data.

Example 2.1. To illustrate, we consider a reduced PBPK model used to predict levels of therapeutic antibody exposure in the brain to guide immunotherapy treatments for central nervous system (CNS) disorders. The original full multispecies platform model [85] is comprised of 100 coupled ordinary differential equations (ODEs). This was later reduced in [55] to provide a minimal PBPK model with 16 ODEs and 37 parameters. This m-PBPK model retains the predictive functionality of the original model but is much simpler and requires approximately 1/11 of the computation time. This significantly facilitates parameter optimization, sensitivity analysis, uncertainty quantification, and real-time diagnoses. This simplification also improves biological interpretability, treatment development, and integration into QSP packages for more comprehensive neurological disease diagnostics.

The 16 compartments and corresponding transport relations for the brain m-PBPK model are illustrated in Figure 2.2 from [55]. For brevity, we summarize a subset of representative model equations and parameters and refer readers to [55] for the complete model.

A balance of antibody concentrations in the plasma, tissue, and lymph compartments yields the coupled ODEs

$$\begin{aligned}
 V_P \cdot \frac{dC_P}{dt} &= (Q_T - L_T) \cdot C_{TV} + (Q_B - L_B) \cdot C_{BV} + (L_T + L_b) \cdot C_L \\
 &\quad - Q_T \cdot C_P - Q_B \cdot C_P, \\
 V_{TV} \cdot \frac{dC_{TV}}{dt} &= Q_T \cdot C_P - (Q_T - L_T) \cdot C_{TV} - (1 - \sigma_{TV}) \cdot L_T C_{TV} \\
 &\quad - CL_{UP_T} \cdot C_{TV} + CL_{UP_T} \cdot F_R \cdot C_{TE,B}, \\
 V_{TE} \cdot \frac{dC_{TE,U}}{dt} &= CL_{UP_T} \cdot (C_{TV} + C_{TI}) - V_{TE} \cdot (k_{onFcRn} \cdot C_{TE,U} \cdot C_{FcRnLi} \\
 &\quad + k_{offFcRn} \cdot C_{TE,B} - k_{deg} \cdot C_{TE,U}), \\
 V_{TE} \cdot \frac{dC_{TE,B}}{dt} &= V_{TE} \cdot (k_{onFcRn} \cdot C_{TE,U} \cdot C_{FcRnLi} - k_{offFcRn} \cdot C_{E,B}) \\
 &\quad - CL_{UP_T} \cdot C_{E,B}, \\
 V_{TI} \cdot \frac{dC_{TI}}{dt} &= (1 - \sigma_{TV}) \cdot L_T \cdot C_{TV} - (1 - C_{TL}) \cdot L_T \cdot C_{TI} \\
 &\quad + CL_{UP_T} \cdot (1 - F_R) \cdot C_{E,B} - CL_{UP_T} \cdot C_{TI}, \\
 V_L \cdot \frac{dC_L}{dt} &= (1 - \sigma_{TL}) \cdot L_T \cdot C_{TI} + (1 - \sigma_{BCSF}) \cdot Q_{BCSF} \cdot C_{BCSF} \\
 &\quad + (1 - \sigma_{BCSF}) Q_{BECR} \cdot C_{BI} - (L_T + L_B) \cdot C_L.
 \end{aligned} \tag{2.6}$$

Here $C_P, C_{TV}, C_{TE,U}, C_{TE,B}$, and C_L denote antibody concentrations in plasma, vascular tissue, unbound and bound endosome tissue, and lymph with units of antibody/L. Moreover, V_P, V_{TV}, V_{TE}, V_L are tissue volumes (L), $Q_T, Q_B, Q_{BCSF}, Q_{BECR}$ denote flow rates (L/h), and L_T, L_E denote lymphatic flow rates (L/h). We summarize a subset of parameters for (2.6), and their units, in Table 2.1. We refer readers to Table 1 of [55] for the full set of 37 model parameters, with representative values for values for four species including humans.

One objective is to employ predicted antibody concentrations in the plasma and brain to evaluate the efficacy of various drug regimes to treat CNS disorders.

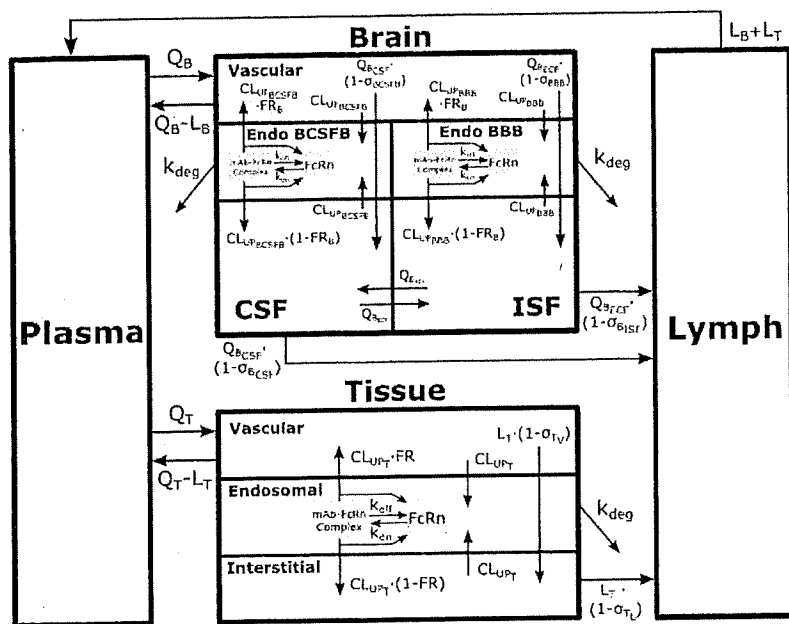


Figure 2.2. Brain m-PBPK compartment model; figure from [55], distributed under CC BY 4.0 DEED License (<https://creativecommons.org/licenses/by/4.0/>).

A specific goal is to bioengineer crystallizable fragment (F_c) regions to extend the half-life of antibodies to increase drug efficiency and decrease the frequency of treatments.

This model encapsulates a number of properties and challenges exhibited by both PBPK and QSP models and, more generally, ODE models for biological systems, which are derived via compartmental analysis.

Biological Component (Units)	Parameters
Tissue Volumes (L)	V_P, V_{TV}, V_{TE}, V_L
Flow Rates (L/h)	Q_T, Q_B, Q_{BCSF}
Lymphatic Flow Rates (L/h)	L_T, L_B
Reflection Coefficients (·)	$\sigma_{TV}, \sigma_{TL}, \sigma_{BCSF}$
Tissue Update Clearance (L/h)	CL_{UP_T}
Antibody – $F_c R_n$ Association Rate (1/h)	$k_{off F_c R_n}$
– $F_c R_n$ Disassociation Rate (1/M/h)	$k_{on F_c R_n}$
Fraction of Antibody Recycled to Plasma Space (·)	FR
Abbreviations: P (Plasma), T_V (Tissue Vascular), T_L (Tissue Lymph), T_E (Tissue Endosome), B (Brain), $BCSF$ (Brain CSF)	

Table 2.1. Subset of parameters in coupled antibody model (2.6).

2.2.1 Properties of PBPK and QSP Models

Physiologically-based pharmacokinetic (PBPK) and, more generally, quantitative systems pharmacology (QSP) models share the following properties, which motivate aspects of sensitivity analysis and uncertainty quantification.

- The models are coupled, nonlinear, and can have a moderate to very large number of states, e.g., 100 for the full PBPK model noted in Example 2.1. They can also have a large number of phenomenological parameters, which often cannot be directly measured from experimental data. Furthermore, the qualitative behavior of models can vary widely in different regions of the parameter space.
- Systems often have components which are far from equilibrium and can produce bifurcations in certain operating regimes. Feedback mechanisms are often nonlinear and highly complex.
- Model parameters typically vary among individuals in a population; e.g., values for the m-PBPK parameters in Table 1 of [55] are patient specific. This must be addressed when constructing virtual populations in the manner detailed in Section 2.6.
- Many states generally cannot be experimentally measured, and data is often limited and exhibits significant noise and uncertainties. For example, one cannot measure all 16 states for the m-PBPK model and data is limited to certain discrete times.
- There are often more parameters than data, which yields underdetermined systems when performing optimization and statistical parameter inference.
- Phenomena often span a range of temporal and spatial scales. This can yield a hierarchy of models with the specific choice of model dictated by the objectives.

2.2.2 Roles of Sensitivity Analysis and Uncertainty Quantification

The m-PBPK model motivates the following issues pertaining to sensitivity analysis, uncertainty quantification, and model reduction. To illustrate the workflow, we indicate chapters where the general topics are addressed. However, we do not provide examples illustrating these techniques for the m-PBPK model. Aspects of this analysis constitute present and future research.

- Model reduction often constitutes a critical initial step prior to sensitivity analysis and uncertainty quantification. This is illustrated by the m-PBPK model, which was constructed by reducing biological complexity while retaining physiological functionality. This provided a reduction from 100 ODEs to 16

2.3. Weather Models

ODEs and 37 parameters. The reduced simulation times support sampling-based sensitivity and uncertainty analysis whereas the reduced complexity improves the biological interpretability of the model. Such models are often termed *reduced-biology* or *reduced-physics* models. Their development is typically application-dependent, which significantly improves functionality but limits extension to other problems. As an alternative, we discuss more generic techniques for surrogate and reduced-order model development in Chapters 15–19.

- The model has parameters, such as the tissue flow rate Q_T , which cannot be uniquely estimated using selected response concentrations, e.g., C_p . The parameter identifiability analysis in Chapter 7 provides conditions which determine if parameters can be uniquely inferred from data. The sensitivity analysis, parameter subset selection, and active subspace analysis in Chapters 8–10 provide techniques to determine subsets or subspaces of influential or identifiable parameters.
- Whereas the model can be employed for a population, parameter values are specific to individual patients. Mixed-effects models, discussed in Section 6.3, provide a framework for such analysis by incorporating both individual and population effects.
- The model parameters are uncertain due to uncertainties and errors in the data. The frequentist and Bayesian techniques in Chapters 11 and 12 provide frameworks to quantify parameter uncertainties via statistical concepts, which include confidence and credible intervals.
- To evaluate the efficacy of drug regimens and determine effective treatment strategies, it is critical to establish the validity and uncertainty of states such as the lymph and brain concentrations C_L and C_{BV} . The uncertainty propagation techniques in Chapter 13 provide a general framework for such analysis.

We discuss the use of QSP models to develop virtual populations in Section 2.6. This is achieved by determining model outputs, which represent desired characteristics of a population in the sense that their distribution matches observed data. For the m-PBPK model, one might construct a virtual population based on a specific antibody concentration. The objective is to use the response of the virtual population to guide drug and treatment discovery in an efficient, safe, and inexpensive manner. This does not replace clinical trials, but rather augments their efficiency and impact.

2.3 Weather Models

Figure 2.3 illustrates some of the physical processes which influence weather and must be incorporated in models. Temperature in the atmosphere depends on the absorption and emission of solar and terrestrial radiation in combination with advection due to winds and convective heating and cooling at the earth's surface.

Temperature fields are also influenced by latent heat produced by phase transitions between liquid, vapor, and solid phases of atmospheric moisture.

Variations in temperature produce pressure gradients, which generate local wind patterns and more global phenomena such as the jet stream. Near the surface of the earth, wind flow is strongly affected by the terrain as well as ground, water, and ice temperatures. High-altitude wind fields, such as the jet stream, tend to be more stable than surface wind patterns, but they still exhibit turbulent dynamics, instabilities, and bifurcations due to the highly nonlinear and complex coupling between atmospheric conditions.

Aerosol levels affect a number of atmospheric and weather processes, including temperature fields, cloud formation, and precipitation. Aerosols are broadly comprised of a range of particulates including products from chemical processes, wind-driven dust particles, smog, combustion byproducts, and residue from volcanic reactions. Aerosols act as a catalyst for phenomena such as cloud formation and precipitation by serving as condensation nuclei which promote the formation of cloud droplets.

2.3.1 Conservation Relations

The physical components of meteorological or weather models are based on conservation of mass, momentum, and energy in combination with conservation of water phases m_i and aerosol concentrations χ_j . As detailed in [289, 319], this yields the coupled set of equations

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) &= 0, \\ \frac{\partial \mathbf{v}}{\partial t} &= -\mathbf{v} \cdot \nabla \mathbf{v} - \frac{1}{\rho} \nabla p - g \hat{\mathbf{k}} - 2\boldsymbol{\Omega} \times \mathbf{v}, \\ \rho c_v \frac{\partial T}{\partial t} + p \nabla \cdot \mathbf{v} &= -\nabla \cdot \mathbf{F} + \nabla \cdot (k \nabla T) + \rho \dot{q}(T, p, \rho), \\ p &= \rho R T, \\ \frac{\partial m_j}{\partial t} &= -\mathbf{v} \cdot \nabla m_j + S_{m_j}(T, m_j, \chi_j, \rho), \quad j = 1, 2, 3, \\ \frac{\partial \chi_j}{\partial t} &= -\mathbf{v} \cdot \nabla \chi_j + S_{\chi_j}(T, \chi_j, \rho), \quad j = 1, \dots, J.\end{aligned}\tag{2.7}$$

These are often referred to as the *equations of atmospheric physics*. The states ρ , $\mathbf{v}$, and T are density, velocity, and temperature whereas m_1 , m_2 , and m_3 denote the concentrations of solid, liquid, and gaseous water phases. The source terms $\mathbf{F}$ and $\dot{q}(T, p, \rho)$ respectively denote the net radiative flux and rate of internal heating or cooling. The source or sink terms S_{m_j} quantify the processes that govern phase transitions of water whereas the source terms S_{χ_j} incorporate changes in state, chemical reactions, and sedimentation, which influence aerosol concentrations.

Various meteorological and climate models are constructed by employing simplified forms of these relations. We note that meteorological models for tropical

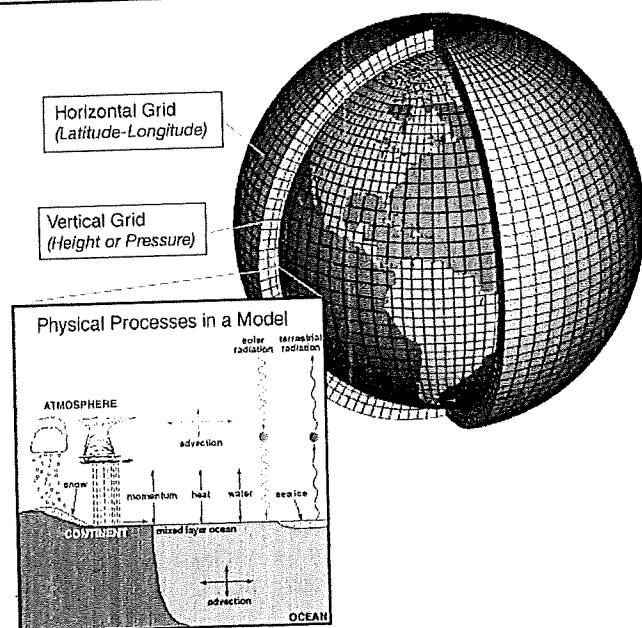


Figure 2.3. Physical processes which must be incorporated in weather and climate models and the associated 3-D grid. Image courtesy of the National Oceanic and Atmospheric Administration.

dynamics are significantly more complex than those for midlatitude or extratropical regions, due to a variety of physical phenomena. We refer readers to Chapter 11 of [178], and the included references, for more details regarding tropical weather phenomena and associated modeling issues.

2.3.2 Phenomenological Models

To solve (2.7), one must provide expressions for the net radiative flux F , the rate $\dot{q}(T, p, \rho)$ due to latent heat release, and the source terms $S_{m_j}(T, m_j, \chi_j, \rho)$ and $S_{\chi_j}(T, \chi_j, \rho)$. One can construct relations for F and $\dot{q}$ based on physical principles. In contrast, the source terms S_{m_j} and S_{χ_j} are phenomenological and occur on scales which are often significantly smaller than computational grids. For these terms, one constructs phenomenological models having parameters which must be estimated using data.

Letting m_2 denote the concentration of liquid phase water, it is established in [289] that one can represent the source term S_{m_2} as

$$S_{m_2} = S_1 + S_2 + S_3 - S_4,$$

where S_1 and S_2 represent conversions between cloud water and raindrops, and S_3 and S_4 quantify the melting of ice and evaporation of rain. A phenomenological

relation for S_1 is

$$S_1 = \bar{\rho}(m_2 - m_2^*)^2 \left[1.2 \times 10^{-4} + \left(1.569 \times 10^{-12} \frac{n_r}{d_0(m_2 - m_2^*)} \right) \right]^{-1}. \quad (2.8)$$

Here $\bar{\rho}$, m_2^* , n_r and d_0 are constants which must be specified or inferred through a fit to data.

The relation (2.8) constitutes one of several phenomenological constitutive and closure relations having parameters which must be inferred before the models can be employed to predict future weather events.

2.3.3 Uncertainty Analysis

Sources of Uncertainty

We noted in Chapter 1 that sources of uncertainty for mechanistic models occur in the model formulation, parameters including initial conditions, numerical solution techniques, and experimental data. We briefly summarize these sources to motivate roles of uncertainty quantification for weather models.

- Sources of uncertainty in the coupled model (2.7) include the formulation of phenomenological relations for forcing terms such as the source/sink terms S_{m_j} and S_{x_j} . These epistemic uncertainties – see Definition 1.11 – introduce model errors or model discrepancy, which has the ramifications discussed in Chapter 14.
- We illustrated in (2.8) the manner in which phenomenological constitutive relations introduce parameters which must be inferred prior to making future predictions. One must also infer initial conditions for the evolution system (2.7). These uncertainties are often aleatoric as defined in Definition 1.10.
- The large 3-D grids employed for weather prediction introduce significant uncertainty when quantifying subgrid phenomena such as turbulence or aerosol levels. For example, one cannot specify processes, which occur on the scale of meters to a few kilometers, using the grid depicted in Figure 2.3, which has horizontal scales on the order of hundreds of kilometers. Second, one incurs discretization errors when approximating the coupled system (2.7). Both sets of errors are epistemic in nature with discretization errors generally quantified using numerical analysis associated with the employed finite-difference, finite-element, spectral, or semi-Lagrangian techniques.
- Variability in the location of sensors can yield epistemic uncertainties in measured data. Many sensors also have limited resolution which can produce aleatoric uncertainties in data.

Ensemble Forecasts and Uncertainty Bounds

In addition to the noted sources of uncertainty, weather models generally exhibit chaotic behavior due to their highly nonlinear nature. Both phenomena are presently addressed by *ensemble* forecasts.

To construct ensemble forecasts using a single model, one samples parameters and initial conditions from probability densities inferred during model calibration and propagates these values through models to predict statistical quantities of interest, such as average temperatures or rainfall. The accuracy of predictions is further increased when multiple models are used to compute *multimodel ensemble forecasts*. To illustrate, a 75% change of rain means that based on the current atmospheric conditions, 3/4 of simulations predict measurable rain amounts at randomly chosen locations in the region. The European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecast System provides such ensemble forecasts for locations around the world.

In addition to predictions, ensemble forecasts can be used to construct uncertainty bounds. We illustrate in Figures 2.4 and 2.5 ensemble forecasts and the cone of uncertainty for the hurricane Katrina. Figure 2.4(b) illustrates the reduction in ensemble variability for predictions which are 12 hours later than those plotted in Figure 2.4(a). The cone of uncertainty was based on historical data for similar conditions. We note that uncertainty bounds computed from ensemble forecasts are typically reported only for large storms such as tropical storms or hurricanes.

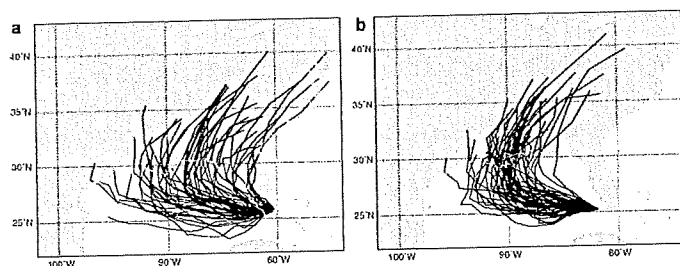


Figure 2.4. ECMWF ensemble forecasts made at (a) 00 UTC and (b) 12 UTC on August 26, 2005. Katrina made landfall near New Orleans at 12 UTC on August 29, 2005. © 2005 European Centre for Medium-Range Weather Forecasts (ECMWF).

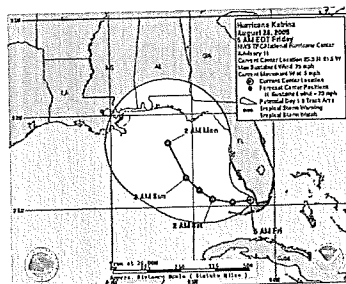


Figure 2.5. NOAA image of the trajectory and cone of uncertainty for Katrina. Image courtesy of the National Oceanic and Atmospheric Administration.

2.3.4 References

We refer readers to [34, 256, 293] for details regarding weather phenomena and [191, 289, 319] for a derivation of the equations of atmospheric physics and associated constitutive relations. Details regarding ensemble forecasts are provided in [196, 221]. Finally, readers can find details regarding data assimilation algorithms, including 4D-VAR, in [196, 268], with an overview of data assimilation for weather models provided in the Supplemental Material.

2.4 Climate Models

Whereas climate and weather models both require the quantification of coupled atmospheric, land, ocean, and solar radiation processes, the vastly differing timescales dictate that different physical mechanisms be emphasized in the two regimes.

We illustrated in Section 2.3.1 that the incorporation of atmospheric physics and chemistry associated with temperature gradients, wind patterns, aerosols, and concentrations of liquid, gaseous, and solid water phases yields the equations of atmospheric physics (2.7). This initial value system is highly nonlinear and exhibits chaotic dynamics which, in part, motivates ensemble predictions. The accuracy of weather models is generally limited to 10 to 14 days with shorter accuracy in the tropics.

In contrast, climate models focus on the long-term balance between natural and anthropogenic forcing effects – such as volcanic ash and increased greenhouse gas levels – and absorbed solar energy and surface radiation reflected back to space. This is termed the earth's *energy budget* or *global energy balance* and it includes effects including those depicted in Figure 2.6. In addition to volcanic emissions and greenhouse gas levels, the energy budget includes effects due to solar radiation, deforestation and urbanization, aerosol levels, polar ice coverage, and global behavior of ocean currents. The focus on this long-term energy balance eliminates the short-

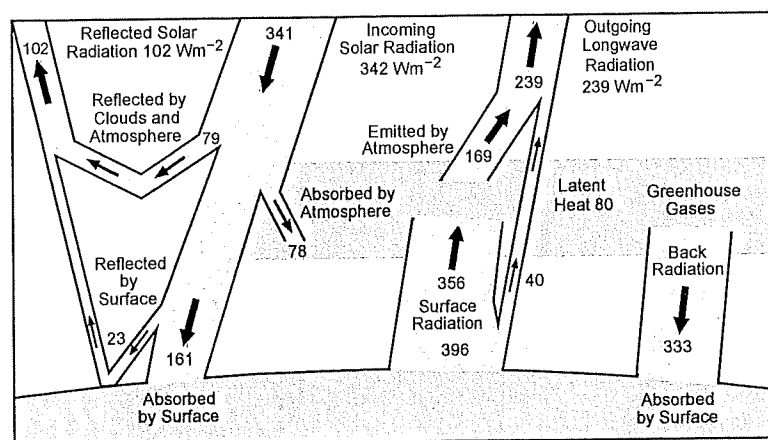


Figure 2.6. Earth's energy budget, modified from [361].

term transient and chaotic behavior of weather models. Instead, climate models reflect the longer-term dynamic behavior of forced boundary value problems.

In contrast to weather, which reflects the state of the coupled atmospheric system on shorter timescales, *climate* refers to the mean state of the climate system. As noted in [197], the climate system is broadly comprised of five components: the atmosphere, the hydrosphere (lakes, oceans, and bodies of water), the lithosphere (land surface), the cryosphere (snow and ice), and the biosphere (living organisms). The term *climate change* denotes the manner in which the statistics of weather change over time.

Some questions posed for climate models include the following.

- Is the earth getting warmer and are manmade processes contributing to the warming?
- Are atmospheric and/or oceanic circulation patterns or currents changing and, if so, what effect will this have on the global energy balance?
- Is weather becoming more extreme and, if so, what ramifications does this have on the climate?

Answers to these questions, with quantified uncertainties, are required to address societal questions.

- Will climate changes lead to changes in agriculture and food supplies or deficits due to widespread drought?
- Will sea-level changes threaten large civilization centers?
- Will changes in ozone levels significantly increase the incidence of cancer?

We note that the Community Earth System Model (CESM) climate package is publicly available and includes atmosphere, land, sea and land ice, and ocean components. We also refer readers to the World Climate Research Programme's Coupled Model Intercomparison Project (CMIP) for resources regarding models and data employed by the Intergovernmental Panel on Climate Change (IPCC) and other international climate assessment organizations.

We next summarize certain climate forces and note sources of uncertainty.

2.4.1 Climate Forces

There are a number of natural and human-induced forcing mechanisms which act on the earth's energy balance to produce changes in climate. There are also numerous processes for which changes in the climate state act as forces which subsequently produce additional climate changes. Termed *feedback processes*, these include ice albedo effects, which is due to the fact that ice and snow reflect more solar energy than open ocean water. This produces a positive feedback mechanism since increasing temperatures reduce the surface area covered by snow and ice, which subsequently produces additional temperature increases.

Natural Forcing Mechanisms

Natural forces include internal or terrestrial events, such as volcanic eruptions, and external forcing mechanisms, including fluctuation in solar radiation. To illustrate the former, we note that aerosols produced by the 1991 eruption of Mount Pinatubo reduced the solar energy reaching the earth's surface by $3\text{--}4\text{ W/m}^2$ and cooled global temperatures by approximately 0.5°C . This occurred after the advent of a global monitoring network so it provided a unique opportunity to incorporate volcanic inputs to climate models. The climate changes due to volcanoes are relatively short-term unless they occur in conjunction with other climate forces.

Whereas we cannot control natural climate forces, it is necessary to quantify them to determine their relative impact compared with anthropogenic forces, which we can influence.

Human-Induced Forcing Mechanisms

Anthropogenic climate forces are those due to human activities. The understanding and quantification of uncertainties associated with anthropogenic forces are of fundamental importance in climate science to establish the degree to which we can influence and, in some cases, control them. We briefly summarize the effects of greenhouse gas emissions, aerosol emissions, and deforestation and urbanization. Readers can find additional details regarding human-induced forcing mechanisms in the references cited in Section 2.4.3.

Greenhouse Gas Emissions

Gases in the atmosphere – including water vapor, CO_2 , methane, and chlorofluorocarbons (CFCs) such as the refrigerant Freon – provide a climate forcing mechanism since they absorb and reflect thermal radiation from the earth's surface. Whereas several of these greenhouse gases occur naturally, levels for a number of the gases have increased substantially due to human activities. As a result, warming due to the greenhouse effect now constitutes a heavily investigated area in anthropogenic climate forcing.

Concentration levels for carbon dioxide, CO_2 , have been extensively monitored and investigated since it is the most abundant greenhouse gas after water vapor and it is a byproduct of burning fossil fuels and internal combustion engines. Figure 2.7 illustrates CO_2 concentrations monitored at Mauna Loa, Hawaii, since 1958 along with CO_2 and N_2O concentrations based on ice core data. Carbon isotopes establish that the increase in CO_2 , since the mid-nineteenth century, is largely influenced by emissions from fossil fuels, which started during the Industrial Revolution.

The effect of certain emissions on radiative forcing and temperature, along with 5%–95% error bars, is illustrated in Figure 2.8 from [17]. This shows the significant role of CO_2 on radiative forcing and changes in global surface temperatures.

Finally, Figure 2.9, from the 2021 IPCC report [17], illustrates observed global surface temperatures, along with reconstructed temperatures from proxy measurements, and values simulated using climate models. It also includes model

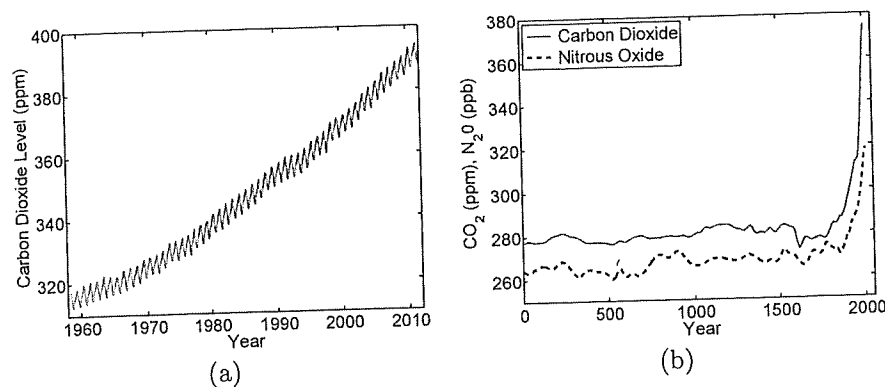


Figure 2.7. (a) Concentration of CO_2 measured at Mauna Loa; data from <https://gml.noaa.gov/ccgg/trends/>. (b) Concentrations of CO_2 and N_2O over the last 2000 years; data from the 2007 IPCC report [336].

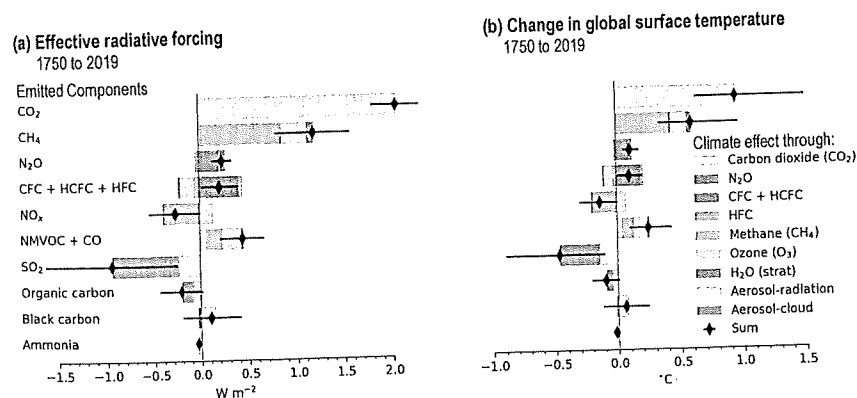


Figure 2.8. Contributions of component emissions, with 5%–95% error bars, to (a) effective radiative forcing and (b) global surface temperature change, based on Coupled Model Intercomparison Project Phase 6 (CMIP) models; Figure TS.15 from the 2021 IPCC Report [17]. Figure TS.15 in IPCC, 2021: Technical Summary. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [Chen, D., M. Rojas, B.H. Samset, K. Cobb, A. Diongue Niang, P. Edwards, S. Emori, S.H. Faria, E. Hawkins, P. Hope, P. Huybrechts, M. Meinshausen, S.K. Mustafa, G.-K. Plattner, and A.-M. Tréguier, 2021: Framing, Context, and Methods. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp. 147–286, <https://doi.org/10.1017/9781009157896.003>.].

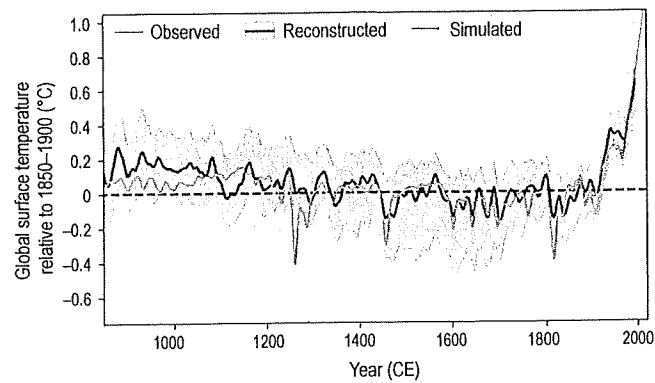


Figure 2.9. Observed, reconstructed, and simulated global surface temperatures with computed model uncertainties; from the 2021 IPCC Report [17]. Box TS.2, Figure 2 in IPCC, 2021: Technical Summary. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [Chen, D., M. Rojas, B.H. Samset, K. Cobb, A. Diongue Niang, P. Edwards, S. Emori, S.H. Faria, E. Hawkins, P. Hope, P. Huybrechts, M. Meinshausen, S.K. Mustafa, G.-K. Plattner, and A.-M. Tréguier, 2021: Framing, Context, and Methods. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp. 147–286, <https://doi.org/10.1017/9781009157896.003>.].

uncertainties based on 5%–95% ranges of multimodel ensemble means. It is noted in the report that improved models and a better understanding of data have led to a better agreement between models and reconstructions as compared to previous IPCC reports. In combination, data, such as that in Figures 2.7–2.9, has been cited by scientists and policy makers as evidence that increasing anthropogenic CO₂ levels are leading to temperature increases which in turn produce global warming.

Aerosol Emissions

We noted in Section 2.3 that aerosols serve as a catalyst for atmospheric phenomena such as cloud formation and precipitation. Although particulates such as dust and volcanic ash arise from natural sources, the substantial growth in aerosol levels due to phenomena such as pollution have caused this to become a substantial human-induced climate forcing mechanism. We note that phenomenological force terms such as S_{x_j} in (2.7) constitute a source of model and parameter uncertainty. Like greenhouse gas levels, aerosol predictions also rely on socioeconomic growth models which are addressed in IPCC reports via the use of scenarios.

Deforestation and Urbanization

The earth's surface vegetation contributes significantly to the global energy balance by absorbing CO₂ via photosynthesis and absorbing solar energy. Approximately 30% of the land surface is presently forested and 11% is arable. These percentages are decreasing, however, as forests are cut and croplands are replaced by urban environments. As with greenhouse gases and aerosols, models quantifying future deforestation and urbanization are often uncertain since they depend, in part, on demographic and socioeconomic factors.

2.4.2 Uncertainty Analysis

Sources of Uncertainty

We summarize here selected sources of uncertainty to motivate issues which must be addressed when employing models to predict future climate trends. This discussion is necessarily brief, and we refer readers to references cited in Section 2.4.3 for additional roles pertaining to uncertainty analysis in climate science.

- As noted in Section 2.4.1, models for external and internal forcing mechanisms are generally phenomenological with parameters which must be estimated using data. This includes terms such as the forcing inputs S_{m_j} and S_{x_j} in models which quantify water phases and aerosols. Moreover, these models are highly coupled and often span significant time and spatial scales. The variation in temporal and spatial scales introduces significant uncertainties in both the models and data. In some cases, the effects of parameter uncertainties are investigated using ensemble predictions based on parameter values sampled from accepted distributions.
- As with weather models, the computation grids employed for numerical simulation packages are often significantly larger than the scales associated with the physical phenomenon, e.g., aerosol-induced cloud formation. This introduces epistemic uncertainties in the model predictions.
- Data is often sparse, incomplete, and lacking uncertainty bounds, often due to its historic nature. For example, ice core and tree ring analyses, used to establish past CO₂ and temperatures, exhibit varying degrees of uncertainty or inaccuracy. Moreover, the designation of which data is utilized and which is neglected has introduced uncertainty into both present interpretations and future predictions.
- The prediction of future CO₂ levels, greenhouse gas effects, and aerosol levels is very difficult and prone to uncertainties since it relies on socioeconomic growth models. This is typically addressed by considering various growth scenarios. The specific designation of scenarios has evolved in IPCC reports, and the 2021 report includes projections based on shared socioeconomic pathways (SSPs) ranging from low to high greenhouse gas emissions. For example, Figure TS1 on page 44 of the 2021 IPCC Report [17] illustrates projected CO₂

and temperature levels, computed using earth system model emulators, for SSP scenarios representing low, moderate, and high emission levels through the year 2300. Whereas these scenarios yield vastly different predictions, they provide potential bounds on long-term climate scenarios.

Roles of Uncertainty Quantification

The central role of uncertainty analysis has long been recognized by the IPCC, and the 2007 report [283] lists several key uncertainties as well as robust findings. Representative uncertainties from that report include the following:

- “Aerosol impacts on the magnitude of the temperature response, on clouds and on precipitation remain uncertain.”
- “Models differ considerably in their estimates of the strength of different feedbacks in the climate system, particularly cloud feedbacks, oceanic heat uptake and carbon cycle feedbacks, although progress has been made in these areas. Also, the confidence in projections is higher for some variables (e.g., temperature) than for others (e.g., precipitation), and it is higher for large spatial scales and longer time average periods.”
- “Large-scale ocean circulation changes beyond the 21st century cannot be reliably assessed because of uncertainties in the meltwater supply from the Greenland ice sheet and model response to the warming.”
- “Projections of climate change and its impacts beyond 2050 are strongly scenario- and model-dependent, and improved projections would require improved understanding of sources of uncertainty and enhancements in systematic observation networks.”
- “Understanding of low-probability/high-impact events and the cumulative impacts of sequences of smaller events, which is required for risk-based approaches to decision-making, is generally limited.”

Additionally, the 2014 report [215] includes discussion regarding uncertainties associated with the following abbreviated list of phenomena.

- “Climate responses to greenhouse gas (GHS) emissions and their associated impacts.” This includes uncertainties pertaining to the evolution of greenhouse gas levels and their impact on climatic and ecological processes.
- “Technological systems.” Topics include the future development of new technologies, the degree to which they contribute to greenhouse gas emissions, and perceptions of health and safety risks.
- “Market behavior and regulatory actions.” This includes uncertainties about market factors such as fossil fuel prices and macroeconomics of aggregate behavioral changes.

Whereas the state of climate science has advanced significantly since these reports, the challenges associated with uncertainties of this nature remain. Hence it is critical that physical, biological, and socioeconomic modeling, data collection and analysis, statistical and mathematical analysis, large-scale computing, and uncertainty quantification continue to be investigated in concert to advance the state of climate models so they can inform scientists in a manner that includes quantified uncertainties.

2.4.3 References

We refer readers to Section 2.3.4 for references detailing the underlying physics for weather and atmospheric processes and [118, 197, 198] for discussion regarding mathematical aspects of climate modeling. In combination, the texts [34, 71, 208, 245] provide a broad perspective regarding many facets of climate and climate change including the global energy balance, natural and human-induced causes of climate change, issues associated with proxy measurements obtained using ice core and tree ring data, and consequences and debates pertaining to climate change. We have selectively cited the Intergovernmental Panel on Climate Change (IPCC) Assessment Reports on Climate Change [17, 116, 215, 283, 336]. This is a small subset of the complete set of IPCC reports and does not include the 2023 Synthesis Report, which is still being revised. We refer readers to additional IPCC reports for further discussion regarding sources and ramifications of uncertainties when predicting climate change and its impact.

2.5 Digital Twins

Whereas the role of mathematical and statistical models for scientific and engineering applications is classical, recent technological, theoretical, and computational advances have transformed the manner in which models are being integrated into evaluation and decision processes. From a technological perspective, the widespread availability of inexpensive sensors, increased connectivity between systems, and broad access to mass data storage facilities provide significant quantities of data for a wide range of applications. These include healthcare, smart cities, next generation and renewable energy facilities, manufacturing, and aircraft and automotive systems. Furthermore, many technologies have evolved in complexity to the point where subsystem failure can yield unanticipated and cascading system events, which preclude classical analysis. Theoretical and algorithmic advances include the widespread development and availability of comprehensive numerical packages for simulation, optimization, statistical inference, and uncertainty propagation for complex, coupled systems. Model capabilities have also been significantly augmented by advances in machine learning and data-driven physics-based models as well as the development of algorithms to process massive amounts of data. Finally, burgeoning computational resources provide novel capabilities for real-time simulations, data processing, and time-sensitive predictions for complex systems.

We summarize in this section the development and role of digital twins for a range of engineering and biomedical systems. We discuss the use of virtual popu-

lations to facilitate drug discovery and treatment regimes in Section 2.6. In both cases, we indicate the present and emerging role of sensitivity and identifiability analysis, uncertainty quantification, and reduced-order models. We note that these are brief introductions to complex and quickly evolving topics and we cite details in the references.

There are numerous definitions of digital twin in the literature. The following definition from [9] encapsulates the attributes that we discuss in this section.

Definition 2.2. A digital twin is a set of virtual information constructs that mimic the structure, context, and behavior of an individual/unique physical asset, or group of physical assets, is dynamically updated with data from its physical twin throughout its life cycle, and informs decisions that realize value.

There are three features which differentiate digital twins from static simulation models. The first is that they quantify the properties and behavior of a specific physical device or biological organ or process. Hence they incorporate both individual properties unique to the physical twin and changes that occur during its life cycle. The latter motivates the second feature, which is the requirement that digital twins be continually updated using physical or biological data. Hence they provide a dynamic, or “living” representation of the process rather than a static model, which is constructed once. Finally, by integrating the model and data throughout the life cycle of the physical or biological process, digital twins serve an increasingly central role in evaluation, monitoring, and decision processes.

The initial digital twin concept is often attributed to NASA with Apollo 13 providing an early example. Following an explosion in the spacecraft’s service module, which critically damaged the main engine, NASA Mission Control used the spacecraft data to update simulation models and inform strategies to bring the spacecraft safely back to earth [123, 271].

From its origins in aerospace engineering, the concept has rapidly expanded within the last decade to significantly impact a growing number of applications. These include structural health monitoring and predictions, aircraft and automotive performance optimization, medical treatment and surgery strategies, dynamic energy usage and allocation, autonomous nuclear power plant monitoring and operation, and manufacturing.

We summarize three examples illustrating the use of digital twins for structural health monitoring for unmanned aerial vehicles, design of nearly autonomous systems to monitor flow anomalies in a nuclear power plant, and guidance of cardiovascular diagnosis and treatment strategies. We refer readers to [18, 128, 260, 271] for details regarding the use of digital twins for other applications.

Example 2.3. We illustrate here the use of a digital twin methodology for structural health monitoring and mission updates for an unmanned aerial vehicle (UAV), which accumulates damage during flight. This example is from [199, 200] and we refer readers to these references for details regarding the digital twin formulation and simulation results.

It is common for many digital twin frameworks to combine a mechanistic physical or biological model with a statistical or machine learning model. By incorporating energy properties, constitutive behavior, closure relations, or potential nonlinear mechanisms, the mechanistic model provides physical interpretability and the capability for extrapolatory or out-of-data predictions. The statistical or machine learning component is required for model selection, data analysis, experimental design, or data-driven compensation for model discrepancy.

For this application, the structural properties of a UAV were modeled using a static condensation reduced basis element (SCRBE) method detailed in [183]. Whereas finite-element packages provide an extremely powerful tool for simulating structural and flight behavior, their complexity generally precludes real-time analysis. As discussed in Chapter 19 for general systems, reduced-order models address this by incorporating the relevant physics in a reduced-basis set. The resulting reduced-basis models permit rapid evaluation while retaining the primary physics.

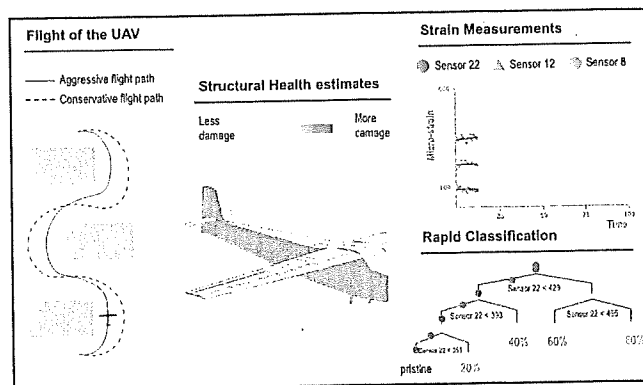
The authors of [200] additionally employed interpretable machine learning to evaluate a library of physics-based damage states for the UAV. This permits the digital twin for the UAV to modify its flight trajectory based on data for 28 damage regions to the wings. The objective was to employ this information for the physical UAV to accommodate accumulating damage.

Figure 2.10 from [200] illustrates the simulated flight trajectory, structural health estimates, strain measurements, and classification trees used to classify the wing damage at two times during the flight. At Timestep 15, the aircraft is undamaged and can withstand aggressive maneuvers with 3g loads. To accommodate accumulated damage, the UAV dynamically changes its flight path by Timestep 75 to reduce loads and avoid potential structural failure. The combined physics and machine learning-based digital twin satisfies the objectives in Definition 2.2 and achieves simulated real-time structural health monitoring and trajectory modifications during flight.

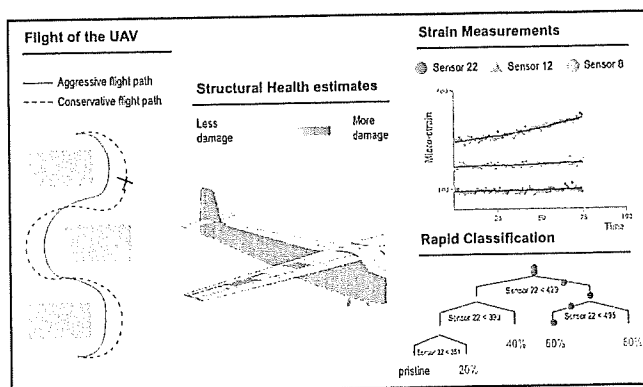
Example 2.4. We illustrated in Section 2.1.3 that the processes in a nuclear power plant involve highly coupled, multiscale physics including neutronics, thermal-hydraulics, chemistry, and fuels. Moreover, the operating conditions often preclude the direct measurement of many processes and states during plant operation, e.g., boron or crud deposits on rods. This highly coupled physics can also produce subsystem anomalies and failures, which are difficult to directly detect yet can significantly impact safe plant operations. This has motivated the use of digital twins for nuclear power plant diagnosis, prognosis, and operation assessment.

We summarize here the use of digital twin technology to develop a nearly autonomous management and control (NAMAC) system for advanced nuclear reactors, as detailed in [229]. This investigation focused on employing digital twins to investigate flow anomalies due to primary sodium pump (PSP) malfunctions. The objective was to evaluate strategies to restore normal flow by activating alternative PSPs or determine if the reactor should be shut down.

The thermal-hydraulic flow dynamics were simulated using the coarse-grid, multiphase computational fluid dynamics (CFD) package GOTHIC coupled with the Risk Analysis and Virtual ENvironment (RAVEN) framework. This provided



(a)



(b)

Figure 2.10. Simulated flight path and damage state at (a) Timestep 15 and (b) Timestep 75; figure from [200]. Reprinted with permission of the authors. Kapteyn, Michael G., David J. Knezevic, and Karen E. Willcox. "Toward predictive digital twins via component-based reduced-order models and interpretable machine learning." *AIAA Scitech 2020 Forum*. 2020.

the flow databases to construct and implement NAMAC. Experimental data was assimilated using feedforward neural networks (FNNs), which were later extended to recurrent neural networks (RNNs). The resulting digital twins were used to predict safety-significant factors, such as peak fuel and cladding temperatures, which were used as dynamically updated surrogates to analyze reactor safety conditions.

To address PSP-induced flow anomalies, the digital twins were employed to investigate available mitigation strategies and predict reactor dynamics. These strategies were subsequently input to plant simulation codes to determine if they mitigated flow anomalies or instead produced sufficiently large flow deviations to require plant shutdown. We again refer readers to [229] for details. We note that this use of digital twins was not meant to replace human plant operators but rather

to provide them with advanced analysis techniques to address unanticipated plant operating scenarios.

Example 2.5. The previous examples illustrate how the objectives of providing dynamically updated simulation models – which quantify individual or group assets of a complex system and are employed for complex decision processes – are producing digital twin technologies for a range of engineering applications. These objectives are also critical in precision medicine, where digital twins are being investigated to provide therapies and surgical options for individual patients. As in engineering applications, biomedical digital twins rely on a combination of mechanistic and statistical models, which are informed and updated in real time using extensive data.

The proposed use of digital twins to enable precision cardiology is detailed in [108]. In this setting, examples of mechanistic models include Navier–Stokes equations for blood flow, bidomain equations for cardiac electrophysiology, and myofiber mechanistic models. Knowledge gained from the data is then assimilated or integrated using statistical or machine learning models, including random forests to assess specific cardiac conditions and unsupervised machine learning to guide patient-specific treatments. The process is illustrated in Figure 2.11.

The paper details several examples where models and digital twins have the potential to significantly impact diagnosis and treatment regimes. For diagnosis, clinical data can be used to infer patient-specific parameters to determine an individual's susceptibility to cardiac conditions including fibrosis or atrial fibrillation. Calibrated models can also be used to infer biomarkers which cannot be directly measured. The paper also cites examples where personalized mechanistic models are presently employed in diagnostic medical devices which have achieved clinical

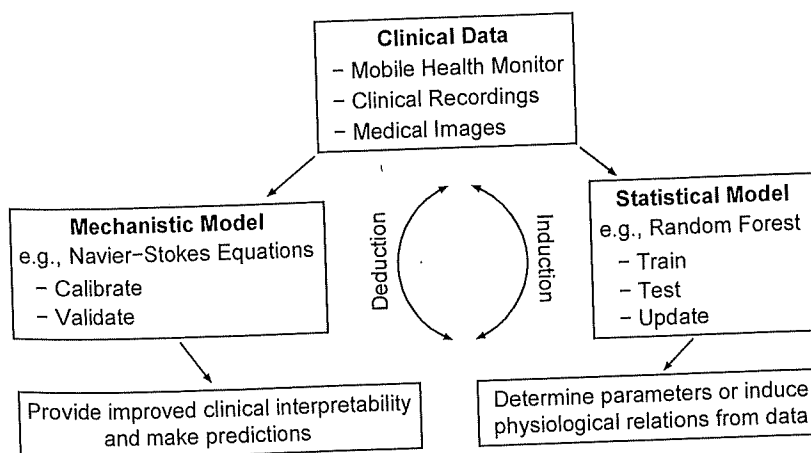


Figure 2.11. Synergy between deduction using mechanistic models and induction using statistical models computed from data to construct digital twins for precision cardiology; based on figure from [108], distributed under CC BY 4.0 DEED License (<https://creativecommons.org/licenses/by/4.0/>).

adoption and FDA approval. This includes HeartFlow FFR_{CT} Analysis, which employs a coronary CT scan to construct a personalized 3-D heart model that can be used to simulate and predict narrowing or stenosis in coronary arteries [108, 254].

Digital twins are also being investigated to guide or optimize treatment strategies for individual patients. One example is the use of data-informed mechanistic simulation models to guide laser ablation to treat atrial fibrillation, which can produce irregular heartbeats.

We refer readers to [18, 128, 271] for additional examples of digital twins in medical, biomedical, and biological applications. Details regarding objectives and perspectives of precision medicine are provided in [210]. In Section 2.6, we discuss virtual populations, which share properties with digital twins, but focus on the use of mechanistic model to guide treatment regimes at the population level.

Roles of Surrogate Models, Sensitivity Analysis, and Uncertainty Quantification

We summarize here some of the roles of surrogate model construction, sensitivity analysis, and uncertainty quantification for digital twins. This list is necessarily abbreviated and will continue to expand as these technologies mature.

- Identifiability or sensitivity analysis is required to determine subsets of model parameters which minimally influence measured responses and must be fixed at nominal values for subsequent optimization and uncertainty analysis. This includes patient-specific parameters for medical diagnosis and treatment, as noted in Example 2.5. Sensitivity analysis can also determine if extensions of models are required to achieve specified functionalities. Finally, sensitivity analysis can be employed to map parameter dependencies across the admissible input space to guide treatment regimes for varying individual traits and demographics.
- Experimental design techniques are often necessary to guide where in the design space to collect future data to best inform mechanistic models. This analysis can complement machine learning when updating models.
- The calibration of mechanistic models requires efficient and robust optimization, frequentist or Bayesian inference, or data assimilation techniques. The statistical techniques are critical for quantifying uncertainties associated with model inputs and experimental observation errors.
- The predictive accuracy of models must be determined and updated to assess potential damage, structural fatigue, risk, or medical feasibility, e.g., accuracy of biomarkers which cannot be directly measured.
- For control applications, quantified model uncertainties can improve controller performance by reducing the effort focused on plant uncertainties to augment tracking capabilities.
- Surrogate or reduced-order models are often required for real-time implementation of digital twins. For applications, such as the UAV damage analysis detailed in Example 2.3, the reduced-order models must retain sufficient physics to permit extrapolatory predictions of future events.

- The impact of uncertainties, when employing digital twins to access risk, inform policy, or increase profits, requires collaboration between the scientists who develop the digital twins and domain experts in the associated fields.

2.6 Virtual Populations

We illustrated in Example 2.1 a minimal physiologically-based pharmacokinetic m-PBPK model comprised of 16 coupled ODE with 37 model parameters. For a given set of parameters, one can compute antibody concentrations in the plasma and brain, which can subsequently be used to evaluate a proposed drug regime. One could potentially employ clinical data for individual patients to construct digital twins to provide treatment regimes. However, this does not facilitate research on drug discovery since predictions are based on a single parameter set.

To employ the quantitative systems pharmacology (QSP) models, discussed in Section 2.2, for drug discovery or the development of broad treatment regimes, it is necessary to employ the models in a manner which reflects attributes of the population under consideration. Virtual populations provide one technique to achieve this objective [10, 313].

To construct a virtual population, it is necessary to first determine appropriate bounds for input parameters and model responses. The parameter bounds may be based on physiological or theoretical constraints. One can additionally impose constraints between parameters, such as known correlation structure, if such information is available. In combination, this provides bounds and constraints for input parameter distributions. For a sufficiently diverse set of population data, it may also be possible to infer parameter distributions using the frequentist or Bayesian techniques discussed in Chapters 11 and 12.

A physiologically viable parameter set, which yields plausible model responses, is often termed a *plausible patient* or *plausible virtual patient*.² We note that if parameters are inferred using specific patient data to create a dynamically updated model, which informs decisions, this would constitute a digital twin in the sense discussed in Section 2.5.

To construct a cohort, one samples a large number of plausible patients, while discarding physiological nonviable candidates, to construct a *plausible population* or *virtual patient cohort*. Whereas this group exhibits a range of attributes, statistical properties of the plausible population may not match those of the considered population, thus limiting its utility for guiding development or clinical research. This is addressed by subsequently optimizing in a manner that improves the statistical agreement between the simulated and observed population data; e.g., matching the mean and moments. Techniques include prevalence weighting [88] or the optimization of inclusion rates [10]. The subset of the plausible population, whose computed response distribution sufficiently matches the experimental distribution of physiological outcomes, is termed a *virtual population*. An individual member is a *virtual patient*. We note that the probability of observing attributes for a virtual patient approximates the probability of similar observations for a physical patient. This

²Terminology can vary among sources when defining plausible patients and populations

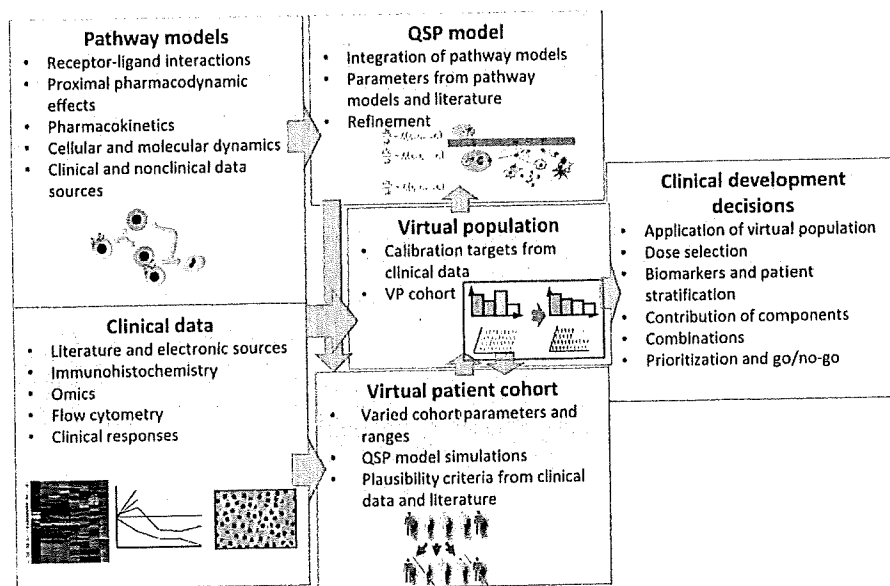


Figure 2.12. Integration of pharmacokinetic processes with clinical data to construct a virtual population; figure from [89], distributed under CC BY 4.0 DEED License (<https://creativecommons.org/licenses/by/4.0/>).

is in contrast to a digital twin, which provides a dynamically updated model for a specific individual.

Figure 2.12 from [89] illustrates the manner in which pharmacokinetic processes, integrated into QSP models, are combined with clinical data to construct a cohort of plausible virtual patients, which are then optimized to construct a virtual population. The virtual population can subsequently be employed to guide clinical decisions.

We summarize in the next example, from [88], the use of a QSP model for tumor drug efficacy and delivery to construct and verify a virtual population. We refer readers to [89] for an example illustrating the construction and validation of a virtual population based on a mechanistic immuno-oncology model, which incorporates a therapeutic response, immune cell dynamics, and lesion response dynamics. Finally, readers can find details regarding the use of virtual populations to guide treatment strategies for COVID-19 in [301].

Example 2.6. Whereas conventional chemotherapy regimens can be highly effective for treating cancer cells, they often damage healthy tissues. To address this issue, antibody drug conjugate (ADC) therapies are being investigated to target tumors using cancer-specific antigens. This reduces the exposure of healthy tissues to the cytotoxic agents, thus improving treatment outcomes. A number of mechanistic ODE models have been developed to guide investigations regarding ADC treatment

protocols. The platform discussed in [88] includes a tumor compartment as well as compartments to simulate drug-to-antibody ratios. This model is comprised of 151 ODEs and was developed in MATLAB SimBiology.

The examples in [88] detail the use of the QSP Toolbox to construct plausible virtual patients (termed virtual xenografts in the paper), a cohort of virtual patients, and virtual populations based on the mechanistic ADC platform. From the perspective of uncertainty quantification, prediction intervals for the virtual population are computed to illustrate the manner in which virtual patients cover parameter ranges. Whereas not discussed, one objective would be to employ the virtual populations to guide subsequent targeted tumor treatments.

Roles of Sensitivity Analysis and Uncertainty Quantification

- Many of the issues detailed in Section 2.2, for the quantitative systems pharmacology (QSP) model, also apply when constructing virtual populations. Moderate- to high-dimensional input spaces necessitate identifiability and sensitivity analysis to determine inputs which directly influence model outputs used to construct and verify a virtual population. Robust statistical inference of model parameters is also critical when constructing plausible patients.
- As noted in Example 2.6, statistical prediction intervals can be employed to quantify the degree to which virtual patients are consistent with parameter ranges.
- In addition to prevalence weighting and optimization of inclusion rates to establish statistical agreement between simulated and observed population data, one area of current research is focused on the use of optimal transport theory to quantify how one probability distribution can be transformed to the other.